

# KHALID AHMAD

Kogi State, Nigeria.

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## Research Interests

Applied Machine Learning, Explainable and Trustworthy AI, Statistical and Deep Learning Modeling, Fairness and Bias in Medical AI, and AI for Healthcare innovation.

## Education

**Federal University, Lokoja**

**Sep. 2018 – Oct. 2023**

*B.Sc. Computer Science, **First Class Honours, CGPA: 4.75/5.0 (95%)***

*Lokoja, Nigeria*

**Class Rank:** 2/159 (Top 1% in the Department, Top 1% Overall)

## Research Experience

**Federal University, Lokoja**

**Oct 2023 – Present**

*Research Assistant*

*Lokoja, Nigeria — Hybrid*

- Reviewing literature for an AI system for early plant disease detection, with emphasis on food safety and nutrition links.
- Built a road anomaly detection system using a DeepConvAutoencoder and Vision Transformer for real-time defect detection.
- Developed a predictive model using academic and non-academic factors to estimate first-year GPA for improved advising.

## Publications

- **Khalid, H.A.**, Fati, O.O, Abdulhakeem A.O, & Aifuobhokhan, J. (Manuscript in Progress). An Interpretable Boosting Framework for Automated Classification of Pattern Electroretinogram Signals with Clinical Metadata.
- Ashiru, A.Z., Mustapha, A., Muhammad, B.A., Aliyu, H.I., & **Khalid, H.A.** (Under review). Discovery of Breast Cancer Biomarkers Using Machine Learning and Systems Biology. *Discover Biotechnology*
- Aifuobhokhan, J., **Khalid, H.A.**, Ekechi, C.C., Olanrewaju, A.O., Afuadajo, E., Bowale, D.A., & Inioluwa, O.M. (2025). Predicting Thyroid Cancer Recurrence Using Machine Learning: An Artificial Intelligence Approach to Clinical Oncology. *International Journal of Advances in Artificial Intelligence and Machine Learning* [DOI].
- Aderemi, I.A., Kehinde, T.O., Ugochukwu, D.O., **Khalid, H.A.**, Adjei, K.Y., & Chijioke, C.E. (2025). Beyond the Black Box: A Systematic Review of Explainable AI for Transparent and Trustworthy Water Quality Monitoring. *IEEE Sensors Reviews* [DOI].
- **Khalid, H.A.**, & Fati, O.O. (2025). A real-time road anomaly detection via masked autoencoder-enhanced vision transformers. *International Journal of Innovative Research in Engineering & Management* [DOI].

## Conferences & Presentations

- **Khalid, H.A.**, Fati, O.O., Faruk, A., & Ismail, A (2025). Machine Learning Models for First-Year GPA Prediction in Nigerian Tertiary Institutions: A Comparative Evaluation. *International Conference on Artificial Intelligence and Robotics (ICAIR)*
- **Khalid, H.A.**. Exploring Technology Hotspots: The Impact of Mathematics on Computer Science. *National Association of Computing Students (NACOS) Workshop, FUL Chapter*, Aug 2022

## Academic & Professional Service

**Deep Learning Indaba**

**Jul 2025 – Aug 2025**

*Judge*

*Kigali, Rwanda — Hybrid*

- Served as a judge for the Deep Learning Indaba 2025 AI Community Challenge in Kigali, contributing to the review of more than 160 teams from over 20 countries and supporting efforts to advance innovative AI research in Africa.

## Teaching Experience

**Digital Onboarders**

**Jan 2024 – Nov 2024**

*CS Mentor & AI Workshop Facilitator*

*Lafia, Nigeria — Hybrid*

- Mentored 78 students in Python, web development, and AI fundamentals through hands-on sessions, workshops, and national events, including the Miva AI Conference.

**Federal University Lokoja**

**Mar 2019 – Jul 2023**

*Academic Tutor*

*Lokoja, Nigeria — Onsite*

- Tutored 2,500+ students in core STEM and computer science courses, improving student performance through personalized instruction and collaborative faculty support.

## Industrial Experience

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### Omdena

Feb 2025 – May 2025

*Machine Learning Engineer*

*New York, USA — Remote*

- Collaborated with a global team of 10 engineers on a Smart Farming project to boost yields for 10,000+ farmers in Kano State, Nigeria.
- Cleaned and engineered features from 148,666 soil moisture records to enhance model inputs, improving data quality and system reliability.
- Trained a Random Forest model that achieved 99% precision in classifying plant health indicators, improving detection accuracy by 9% over baseline.

### Saint Louis University

Jun 2023 – Jul 2023

*Data Analyst*

*Missouri, USA — Remote*

- Analyzed digital marketing campaign data for Finance and Marketing teams, tracking CPC, CPR, reach, and engagement metrics across 10 campaigns.
- Built interactive dashboards using Python, SQL, and Power BI to visualize performance insights, identifying top-performing campaigns (2 and 8).
- Reported ad spend of INR 12,088.61, reaching 188,868 users and generating 5,257 unique link clicks; recommended optimizations for underperformers (3 and 10).

### Nigerian Midstream & Downstream Petroleum Regulatory Authority

Mar 2022 – Aug 2022

*Data Scientist Intern*

*Lokoja, Nigeria — Onsite*

- Analyzed 150+ petrol samples from 40+ Lokoja filling stations, uncovering a 22% adulteration rate and informing regulatory action.
- Built ML models (Logistic Regression, Decision Trees, Random Forest, SVM) with 75% accuracy in detecting fuel adulteration using preprocessed data in Python.
- Reduced manual inspection workload by 30% by integrating predictive analytics into existing regulatory workflows.

### Al-Harir Global Services

Jan 2021 – Jan 2022

*Frontend Developer*

*Lokoja, Nigeria — Hybrid*

- Collaborated with UI/UX designers to build a responsive and visually appealing marketing landing page, improving user engagement.
- Implemented cross-browser responsive layouts using HTML, CSS, and JavaScript, enhancing site accessibility across mobile and desktop platforms.

## Technical Skills

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**Programming & Tools:** Python, C++, JavaScript, HTML, CSS, SQL, Git, Docker, VS Code, Jupyter, Azure

**Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, NLTK, Matplotlib, Hugging Face

**Other Skills:** LaTeX, Microsoft Word, Google Docs, Problem solving, Critical thinking, Team collaboration, Research and documentation, Communication, Time management, Attention to detail, Adaptability

## Honors & Awards

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- Wale Lab Research Internship – 2025: Fully funded AI healthcare research internship.
- Departmental and Faculty Fee Waiver – 2023: Awarded for top academic performance and engagement.
- Ayomide Ajiboye Scholar (USA) – 2023: Scholarship and mentorship for academic excellence.
- Student Tutor of the Year – 2023: Recognized for outstanding peer tutoring.
- Department Contribution Award – 2022: Honored for impactful research and academic service.
- NMDPRA Recommendation – 2022: Commended for exceptional work with Nigeria's petroleum authority.
- Dean's Honor Award – 2020–2023: Multiple-year award for sustained academic excellence.
- Kogi State Bursary Scholar – 2020: Financial award for academic merit and need.

## Volunteering Experience

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### United Nations International Children's Emergency Fund (UNICEF)

Aug 2023 – Dec 2023

*Data Entry Analyst*

*Lokoja, Nigeria — Hybrid*

- Collected and managed population health data from 500+ households and institutions, registering 1,200+ newborns to strengthen UNICEF's local health reporting system.

### Federal University Lokoja

Mar 2021 – Oct 2023

*Social Media Moderator*

*Lokoja, Nigeria — Hybrid*

- Co-managed an 80,000+ member student support network, improving campus communication and assisting students with academic, housing, and registration issues.

## Professional Affiliations

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- Member, Black in AI (BAI)
- Member, Data Science Nigeria (DSN)
- Member, National Association of Computing Students (NACOS)
- Member, Nigeria Computer Society (NCS)
- Core Member, Google Developer Student Clubs (GDSC)